MILITARY SPECIFICATION

TRANSMITTING SET, TELEMETRIC DATA AN/DKT-61(V)

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the manufacture and acceptance of the Telemetric Data Transmitting Set AN/DKT-61(V), herein (see 6.4.4) referred to as the "set".

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated-Plywood

Military

MIL-P-116	Preservation, Methods of
MIL-T-18303	Test Procedures; Preproduction, Acceptance, and Life for Aircraft Electronic Equipment, Format for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, New Jersey 08733, by using the self-addresssed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS -Continued

MIL-M-38510/9	Microcircuits, Digital, TTL, Shift Registers, Monolithic Silicon
MIL-M-38510/163	Microcircuits, Digital, TTL, Hex Bus Drivers with 3-State Outputs, Monolithic Silicon

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-781	Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Report, Preparation of
MIL-STD-2074	Failure Classification for Reliability Testing
MIL-STD-45662	Calibration Systems Requirements

2.1.2 Other Government documents, drawings, and publications. The following other Government drawings and publications form a part of this specification to the extent specified herein.

DRAWINGS

Naval Air Systems Command
(Code Ident 30003)

DL1471AS100	Transmitting Set, Telemetric Data, AN/DKT-61(V)
1471AS219	Shipping and Storage Container

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DRAWINGS -Continued

917AS6626	Signature Characteristics

PUBLICATIONS

Naval Air Systems Command
(Code Ident 30003)

NAVAIRINST 5400.67	Certification program for Navy air-launched guided weapon test systems
WS-6536	Procedures and Requirements for Preparation for Soldering of Electrical Connections

Inter-Range Instrumentation Group (IRIG)

106-80	Telemetry Standards

2.1.2.1 Issues of other Government drawings and publications. Unless otherwise specified herein, the preceding other Government drawings and publications of the issue in effect on date of invitation for bids or requests for proposal form a part of this specification to the extent specified herein.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be otained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Set description. The set shall be composed of mechanical components and electronic circuitry that receives missile information, converts the information to a form compatible with radio frequency (RF) transmission, and transmits the information to receiving sites for recording, display, and analysis.

3.1.1 Major components list. The set shall consist of the following major components:

a. Analog Processor
b. Discrete Processor
c. Missile-Based Computer (MBC) Processor
d. Timing Processor
e. Video Processor

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f. Power Supply
g. Transformer Assembly
h. Mother Board Assembly
i. Extender
j. Transmitter
k. Antenna
l. Shell

3.1.2 Block diagram. Figure 1 illustrates a functional block diagram of the set.

3.1.3 Pulse code modulation (PCM) configuration. The output word configuration of the PCM modulation on the voltage controlled oscillator (VCO) shall be as specified in Table I. The PCM shall contain 128 words per frame, 8 bits per word, and the format shall be nonreturn to zero level (NRZ-L) in accordance with IRIG 106-80. The PCM bit rate shall be 256 kilobits per second (kb/s) when controlled by the missile inputs and 259 kb/s when controlled by the set internal clock.

TABLE I. PCM frame configuration. 1/

2/ A1	A7	A13	A3	A6	A17	A23	A18	A19	A2	A8	A10	D3	MBC 1	MBC 2	3/ A16
A1	MBC 3	MBC 4	A3	MBC 5	MBC 6	A23	MBC 7	MBC 8	A4	MBC 9	MBC 10	A11	MBC 11	MBC 12	A21
A1	MBC 13	MBC 14	A3	MBC 15	MBC 16	A23	MBC 17	MBC 18	A2	A8	A10	A9	MBC 19	MBC 20	A20
A1	MBC 21	MBC 22	A3	MBC 23	MBC 24	A23	MBC 25	MBC 26	A4	MBC 27	MBC 28	A11	MBC 29	MBC 30	D1
A1	MBC 31	MBC 32	A3	MBC 33	MBC 34	A23	MBC 35	MBC 36	A2	A8	A10	A15	MBC 37	MBC 38	A5
A1	MBC 39	MBC 40	A3	MBC 41	MBC 42	A23	MBC 43	MBC 44	A4	MBC 45	MBC 46	A11	MBC 47	MBC 48	A21
A1	MBC 49	MBC 50	A3	MBC 51	MBC 52	A23	MBC 53	MBC 54	A2	A8	A10	A9	MBC 55	MBC 56	A12
A1	MBC 57	MBC 58	A3	MBC 59	MBC 60	A23	MBC 61	MBC 62	A4	D2	A14	A11	4/ Frame Sync		

1/ Word nomenclature is identified in Table IX.
2/ Word number 1
3/ Word number 16
4/ Word numbers 126, 127, and 128. Frame synchronization (Sync) pattern is:
Binary: 1111 1010 1111 0011 0010 0000
Hexidecimal: FAF320

3.2 First article. When specified, a sample shall be subjected to first article inspection (see 4.6.3 and 6.3).

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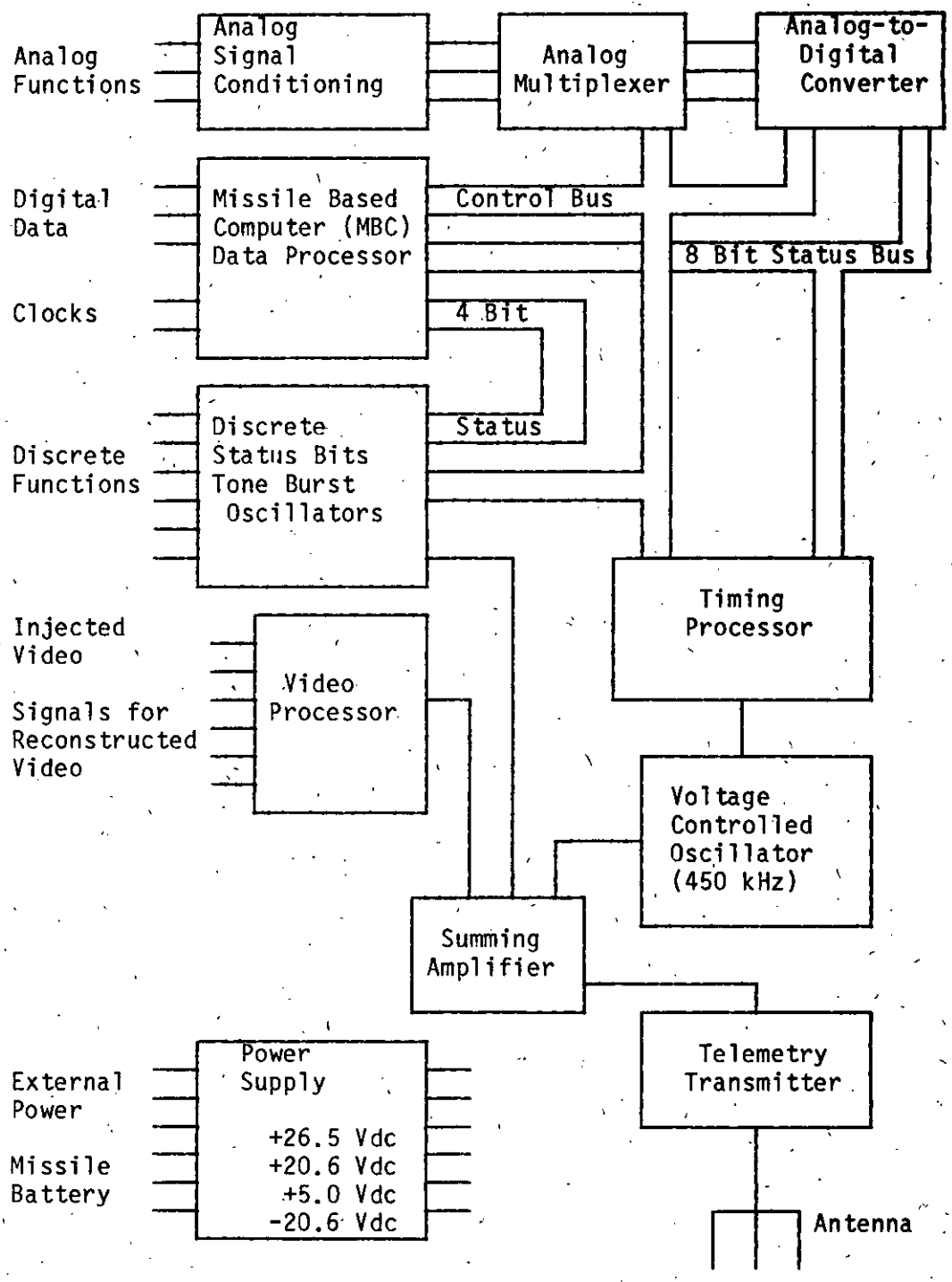


FIGURE 1. Functional block diagram.

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3.2.1 Manufacturing processes and tests. The first article sample of the set shall be manufactured using the same methods, materials, processes, and testing methods proposed for production.

3.2.2 First article sample. The first article sample shall be tested as specified in Section 4 herein for the purpose of determining that, before starting production, the contractor's production methods shall produce sets that meet the requirements of this specification. No production set shall be delivered before approval of the first article sample. Fabrication of production sets prior to such approval is at the contractor's own risk. The first article sample shall not be considered as part of the sets supplied under the contract or purchase order.

3.3 Production set. The set supplied under the contract or purchase order shall in all respects be equal to the approved first article sample. Each set shall be capable of successfully passing the same tests as imposed on the first article sample. Evidence of noncompliance with the above shall constitute cause for rejection.

3.3.1 Production requirements. Following the approval of the first article sample, the contractor shall not change the manufacturing methods, processes, materials, or test procedures used in producing the sample without prior written approval of the procuring activity. Requests for changes shall be by engineering change proposal or request for deviations or waivers (see 6.2.2a). The first article inspection, or any portion thereof, shall be repeated at no additional cost to the procuring activity under any of the following conditions:

a. The contractor has modified the set by a change of sources of supply, production processes or procedures, or test procedures.

b. When there is evidence that the minimum quality requirements of the set have not been maintained. This evidence may be in the form of accumulated failure reports (see 6.2.2j).

c. When production sets fail to meet the requirements of this specification.

3.4 Design and construction. Design and construction of the set shall be in accordance with the drawings and documents listed on DL1471AS100 and as specified herein (see 4.9.7).

3.5 General performance.

3.5.1 Environmental. The set shall provide satisfactory operation (see 6.4.1) before, during, and after any of the environments specified in 3.5.1.1 through 3.5.1.5 and any combination of the environments.

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3.5.1.1 Temperature.

3.5.1.1.1 Storage temperature. The set shall provide satisfactory operation after storage in the shipping and storage container, Drawing 1471AS219, for not less than three years under atmospheric temperature extremes ranging from -54 degrees Celsius (C) to +85C (see 4.8 and 4.8.2).

3.5.1.1.2 Operating temperature. The set shall provide satisfactory operation at temperatures from -40C to +71C (see 4.8 through 4.8.2).

3.5.1.2 Altitude. The set shall provide satisfactory operation from sea level to 90,000 feet altitude (see 4.8 and 4.8.2).

3.5.1.3 Humidity. The set shall provide satisfactory operation under humidity conditions up to 95 percent at all operating temperatures (see 4.8 and 4.8.5).

3.5.1.4 Vibration. When tested in accordance with 4.8.4, the set shall provide satisfactory operation under vibration applied along the longitudinal, vertical, and horizontal axes (see 4.8 and 4.8.4).

3.5.1.5 Shock. The set shall provide satisfactory operation with no damage or maladjustment indicated after impact shocks of 30 gravity units (G) applied along the longitudinal, vertical, and horizontal axes separately (see 4.8 and 4.8.3).

3.5.2 Interchangeability. All assemblies, subassemblies, and parts complying with the requirements of the same drawing number shall be physically and functionally interchangeable (see 4.9.7).

3.5.3 Electromagnetic interference (EMI). The set shall meet the requirements of MIL-STD-461, Methods CE03, RE03; and the Spurious Emission and Interference Requirements of IRIG 106-80 (see 4.9.9).

3.5.4 RF carrier frequency. The RF carrier frequency of each set shall be one of the following: 2200.5, 2212.5, 2228.5, 2236.5, 2244.5, 2252.5, 2264.5, or 2272.5 megahertz (MHz) as specified in the contract or purchase order (see 6.2.1c).

3.6 Detailed performance.

3.6.1 Operating power. The set shall provide satisfactory operation (see 6.4.1) when energized with three phase, 400 hertz (Hz) external power or missile battery power (see 4.9.2).

3.6.1.1 External power. The set shall provide satisfactory operation when energized from simulated aircraft power of 113.5 +5.0 volts root mean square (Vrms), 400 +10 Hz, three phase, 90 volt-amperes (VA) maximum (see 4.9.2 through 4.9.2.1.2).

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3.6.1.2 Missile battery power. The set shall provide satisfactory operation when energized from missile battery power of +26.5 ±3.0 volts direct current (Vdc), 1.0 ampere maximum; +20.6 ±3.0 Vdc, 1.0 ampere maximum; and -20.6 ±3.0 Vdc, 0.3 ampere maximum (see 4.9.2 and 4.9.2.2 through 4.9.2.2.2).

3.6.2 Signal conditioning performance.

3.6.2.1 Analog. The total analog signal conditioning performance shall be within ±6 bits (see 6.4.3) on each analog channel. The total analog signal conditioning performance shall include all combinations of accuracy, linearity, noise, and crosstalk (see 4.9.1 and 4.9.1.1).

3.6.2.1.1 Accuracy and linearity. The accuracy and linearity on each analog channel shall include the effects of gain and offset in the analog signal conditioning and shall be not greater than ±3 bits (see 4.9.1 and 4.9.1.1 through 4.9.1.1.1.2).

3.6.2.1.2 Noise. The noise on each analog channel shall be not greater than ±3 bits (see 4.9.1, 4.9.1.1 and 4.9.1.1.2).

3.6.2.1.3 Crosstalk. The crosstalk on each analog channel from any other analog channel shall be not greater than ±3 bits (see 4.9.1, 4.9.1.1 and 4.9.1.1.3 through 4.9.1.1.3.3).

3.6.2.2 Discrete. The discrete signal conditioning accuracy shall be zero bit errors (see 4.9.1 and 4.9.1.2 through 4.9.1.2.1.2).

3.6.2.3 MBC. The MBC signal conditioning accuracy shall be zero bit errors (see 4.9.1 and 4.9.1.3 through 4.9.1.3.3).

3.6.2.4 Video. The video signal shall consist of injected video and reconstructed video (see 4.9.1 and 4.9.1.4).

3.6.2.4.1 Injected video. With an input between 10 kilohertz (kHz) and 200 kHz of an amplitude of 60 millivolts root mean square (mVrms), the harmonic distortion for the injected video output between 10 kHz and 200 kHz shall be not greater than one percent (see 4.9.1 and 4.9.1.4.1 through 4.9.1.4.1.2).

3.6.2.4.2 Reconstructed video. With an input as specified in Table IX, Item 62, 63, and 64; the output shall be a sine wave with harmonic distortion between 10 kHz and 200 kHz not greater than one percent (see 4.9.1, 4.9.1.4 and 4.9.1.4.2 through 4.9.1.4.2.2).

3.6.2.5 Composite fuse. The composite fuse shall have the following charactaristics specified in 3.6.2.5.1 and 3.6.2.5.2 (see 4.9.1.5).

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3.6.2.5.1 Composite fuse PCM output. The composite fuse PCM outputs shall be as specified in Table IX, Item 24, and shall increase monitonically. The difference between groups shall be not less than six bits (see 4.9.1, 4.9.1.5 and 4.9.1.5.2).

3.6.2.5.2 Composite fuse tone burst outputs. The composite fuse tone burst outputs shall be as specified in Table II (see 4.9.1, 4.9.1.5 and 4.9.1.5.3).

3.6.2.6 Input impedance. The impedance between the input to the set and the return shall be:

Input	Return	Name	Impedance
P35-16	P34-24	HEAD PITCH GYRO	20.5 kilohms ±2%
J35-34	P34-24	HEAD YAW GYRO	20.5 kilohms ±2%
P19-2	P19-1	FUSE OUT PULSE	1.47 ohms ±5%
P1-28	P1-13	LAUNCH +0.6 SEC	not less than 100 kilohms
P35-33	P34-24	SEA SP COMMAND	not less than 100 kilohms

3.6.3 Transmitter. The transmitter shall have the characteristics specified in 3.6.3.1 through 3.6.3.4 (see 4.9.3).

3.6.3.1 RF power output. The RF incident power (see 6.4.5) shall be not less than two watts under all combinations of input power and temperature into any load of any voltage standing wave ratio (VSWR) at any phase angle, referenced to 50 ohms resistive (see 4.9.3.1).

3.6.3.2 RF carrier frequency tolerance. The RF carrier frequency tolerance shall be ±0.003 percent of the assigned frequency under all conditions specified herein (see 4.9.3.2).

3.6.3.3 Modulation. The transmitter modulating signals shall have the characteristics as specified in Table II (see 4.9.3.3).

TABLE II. Modulation characteristics.

Signal	Frequency	Peak transmitter carrier deviation
3.00 kHz Tone Burst	3.00 kHz ± 5%	30.0 kHz ± 5.0 kHz
5.40 kHz Tone Burst	5.40 kHz ± 5%	30.0 kHz ± 5.0 kHz
7.35 kHz Tone Burst	7.35 kHz ± 5%	30.0 kHz ± 5.0 kHz
10.5 kHz Tone Burst	10.5 kHz ± 5%	30.0 kHz ± 5.0 kHz
Injected Video	40 to 100 kHz	155 kHz ± 10 kHz
Reconstructed Video	10 to 200 kHz	155 kHz ± 10 kHz
450 kHz VCO	450 kHz	331 kHz ± 15 kHz

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3.6.3.4 Intermodulation and harmonic distortion. Intermodulation and harmonic distortion products shall be not less than 40 decibels (dB) down from the level of a 40 kHz injected video test signal (see 4.9.3.4).

3.6.4 Transmitter disable. The use of SEA SP COMMAND shall disable the operation of the transmitter after 205 ±5 seconds when operating from missile external power (see 4.9.4 and 4.9.4.1).

3.6.5 Antenna. The antenna shall have a VSWR of not greater than 2.0:1 from 2200 MHz to 2300 MHz under all operating environmental conditions (see 4.9.5).

3.6.6 VCO. The deviation of the VCO by the PCM shall be ±92 kHz with a tolerance of ±10 kHz (see 4.9.6).

3.7 Reliability. The set shall have a mean time between failures (MTBF) of not less than 30 hours (see 4.7.3 and 4.7.3.1).

3.8 Disposition of sets. The disposition of the sets used in reliability demonstration and first article inspection shall be as specified in the contract or purchase order (see 6.2.1f).

3.9 Workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9 (See 4.9.8).

3.9.1 Soldering. Soldering shall be in accordance with WS-6536 (see 4.9.8).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, after approval by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.6)

b. Quality conformance inspection (see 4.7)

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4.3 Documentation.

4.3.1 Test and examination procedures. The contractor shall prepare written detailed test and examination procedures as specified herein in accordance with MIL-T-18303. The procedures shall consist of documents corresponding to the inspections and tests of 4.6 and 4.7. All test and examination procedures shall be subject to acceptance by the procuring activity (see 6.2.2b, g, and h).

4.3.2 Test reports. The contractor shall prepare test reports which document all examinations, first article inspections, and quality conformance inspections. The test reports shall be in accordance with MIL-STD-831. All test reports shall be subject to acceptance by the procuring activity (see 6.2.2c, f, e, and i).

4.4 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in 4.4.1.

4.4.1 Test conditions. The test conditions shall be as follows:

a.	Temperature	23C ±10C
b.	Relative humidity	50 percent ±30 percent
c.	Atmospheric pressure	725 + 50, -75 millimeters of mercury

4.5 Test equipment.

4.5.1 Calibration and maintenance. Unless otherwise specified herein, all test equipment shall be calibrated and maintained in accordance with MIL-STD-45662.

4.5.1.2 Test system certification. Unless otherwise specified in the contract or purchase order (see 6.2.1i), the test systems shall be certified in accordance with NAVAIRINST 5400.67.

4.5.1.3 Test equipment accuracy. The test equipment shall meet the following accuracy requirements:

a. Measurements to meet open-ended maximum or minimum requirements shall be accurate to within 10 percent of the requirement being satisfied.

b. All other measurements shall be accurate to within 10 percent of the tolerance on the requirement to be satisfied.

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4.5.1.4 Input tolerance. Unless otherwise specified herein, all input tolerances shall be as follows:

a. Inputs voltage ranges less than or equal to 1.5 Vdc shall be maintained within ±0.005 Vdc.

b. Inputs voltage ranges greater than 1.5 Vdc and less than or equal to 10.0 Vdc shall be maintained within ±0.010 Vdc.

c. Inputs voltage ranges greater than 10.0 Vdc shall be maintained within ±0.050 Vdc.

4.6 First article inspection. Unless otherwise specified in the contract or purchase order (see 6.2.1h), a first article shall be subjected to the first article inspection of Table III. A first article inspection procedure shall be prepared by the contractor and shall be subject to acceptance by the procuring activity (see 6.2.2b). No adjustments shall be allowed to the sets during or after any tests or examinations. The first article inspection shall be conducted at a test facility approved by the procuring activity (see 6.2.1j). The first article inspection shall be subject to monitoring by the Government.

TABLE III. First article inspection.

Examination or test 1/ 2/	Requirement paragraph	Examination or test paragraph
Workmanship	3.9	4.9.8
Design and construction	3.4	4.9.7
Interchangeability	3.5.2	4.9.7
Temperature-and-vibration conditioning	3.5.1.1.2 and 3.5.1.4	4.8 and 4.8.1
Temperature-altitude	3.5.1.1, 3.5.1.1.2 and 3.5.1.2	4.8 and 4.8.2
Shock	3.5.1.5	4.8 and 4.8.3
Vibration	3.5.1.4	4.8 and 4.8.4
Humidity	3.5.1.3	4.8 and 4.8.5
Electromagnetic interference	3.5.3	4.9.9

1/ The first tests that each first article sample shall be subjected to shall be the workmanship examination, interchangeability examination, design and construction examination, and temperature-and-vibration conditioning test, in that order.

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TABLE III. First article inspection. -Continued

2/a. Following the temperature-and-vibration conditioning test, the first set shall be subjected to the temperature-altitude test. Thereafter, the remaining tests may be performed in any order.

b. Following the temperature-and-vibration conditioning test, the second set shall be subjected to the vibration test. Thereafter, the remaining tests (including the temperature-altitude test) may be performed in any order.

4.6.1 Acceptance criteria. Failure of any set to pass any examination or test of the first article inspection shall result in rejection of the first article sample. Failure reporting and analysis shall be performed by the contractor in accordance with MIL-STD-781, and corrective action shall be proposed prior to disposition of the failed sample (see 3.8 , 6.2.1f and 6.2.2j).

4.6.2 First article approval. Approval of the first article shall be by the procuring activity upon satisfactory completion of the first article inspection. A first article inspection report shall be prepared by the contractor and shall be subject to acceptance by the procuring activity (see 6.2.2c).

4.6.3 First article sample. Unless otherwise specified in the contract or purchase order (see 6.2.1e), the first article sample shall consist of two sets. The first article sample shall be representative of the methods, materials, processes, and test methods the contractor proposes to use in production of the set (see 6.3).

4.7 Quality conformance inspection. The quality conformance inspection shall be performed on each set offered for acceptance, except that the reliability demonstration test shall be performed on only those sets as specified in 4.7.3. The quality conformance inspection shall consist of the examinations and tests of Table IV in the order shown. No adjustments to the set shall be allowed during or after any of the tests or examinations. Failure of any set to pass the quality conformance inspection shall be cause for rejection of the set. Lot acceptance criteria shall be in accordance with 4.7.3. A quality conformance inspection procedure and quality conformance inspection report shall be prepared by the contractor and shall be submitted for acceptance by the procuring activity (see 6.2.2h and i).

4.7.1 Test facility and test monitoring. The set shall be inspected at a test facility approved by the procuring activity (see 6.2.1j). All quality conformance inspection examinations and tests shall be subject to monitoring by the Government inspector.

4.7.2 Inspection lot. Unless otherwise specified in the contract or purchase order (see 6.2.1k), an inspection lot shall consist of all the sets offered to the Government for acceptance in one calendar month.

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TABLE IV. Quality conformance inspection.

Examination or test	Requirement paragraph	Examination or test paragraph
Group A		
Workmanship	3.9	4.9.8
Design and construction	3.4	4.9.7
Interchangeability	3.5.2	4.9.7
Group B		
Temperature-and-vibration conditioning	3.5.1.1.2 3.5.1.4	4.8 and 4.8.1
Group C		
Reliability demonstration 1/	3.7	4.7.3

1/ The reliability demonstration test shall be the last test performed. Three sets shall be selected from each inspection lot (see 4.7.2) and subjected to the reliability demonstration test in accordance with 4.7.3.

4.7.3 Reliability demonstration test. A reliability demonstration test procedure shall be prepared by the contractor in accordance with MIL-STD-781 and shall be subject to acceptance by the procuring activity (see 6.2.2g). A reliability demonstration test shall be conducted in accordance with MIL-STD-781, Test Plan XIXC. The reliability demonstration test environments and performance characteristics shall be in accordance with 4.7.3.1. Only relevant failures shall be used to determine the accept/reject criteria of MIL-STD-781, Test Plan XIXC. Classification of failures shall be in accordance with MIL-STD-2074 (see 4.9.10). Three sets for reliability demonstration shall be randomly selected by the procuring activity representative from each production lot. Each reliability demonstration test set shall have first passed the inspections and tests of 4.8.1. In the event of a reject decision in accordance with MIL-STD-781, Test Plan XIXC, the contractor shall implement corrective action, approved by the procuring activity, to eliminate the deficiencies identified as a result of the reliability demonstration test on all sets which are part of the production lot in question. The Government reserves the right to reject any production lot which does not successfully complete the reliability demonstration test.

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4.7.3.1 Test details. The test level, duty cycle , and performance characteristics to be measured during the reliability demonstration tests shall be as follows:

a. Test levels:

Temperature: -40C to +71C. A programmed temperature chamber shall be used which shall be controlled from the base of the transmitter within +4C of the temperature specified in Figure 2. The dwell time at each temperature shall be two hours.

Vibration: The vibration shall be applied perpendicular to the longitudinal axis of the set and perpendicular to the plane of the printed circuit boards. The vibration level shall be in accordance with 4.8.1.2 except that in Table VI the level of W_1 shall be 0.01 G^2/Hz and the level of W_2 shall be 0.02 G^2/Hz. The vibration time shall be one hour.

Input voltage: The input voltage shall be 113.5 $\pm$ 2.0 Vrms, 400 $\pm$ 10 Hz.

b. Duty cycle: The set may be turned off during the cooling period and shall be energized continuously during the heating portion of the cycle.

c. Performance characteristics: The tests specified in 4.8.1 shall be conducted. The test reports shall be prepared by the contractor and shall be subject to acceptance by the procuring activity (see 6.2.2f).

4.8 Environmental inspections. All environmental tests shall be in accordance with MIL-STD-810, except as specified herein. If a test listed in the examination or test column of Table III or Table IV is conducted within 48 hours of the preceding test's conclusion, it shall not be necessary to conduct the initial performance check on the set.

4.8.1 Temperature-and-vibration conditioning test. The set shall be operated and tested under the conditions specified herein. The set shall be subjected to temperature conditioning first, followed by vibration conditioning. Failures encountered and the corrective action taken shall be recorded and made available for Government inspection. A failure shall be anything which causes malfunctioning of the set. Should a failure occur, the set shall be repaired and the test started over. Should repetitive failures occur, corrective action shall be taken to eliminate the defect from future sets. Each set shall be subjected to the conditioning tests specified in 4.8.1.1 and 4.8.1.2.

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4.8.1.1 Temperature conditioning test. The temperature conditioning test shall be conducted in accordance with Figure 2, Thermal cycle. The following tests shall be performed at the times indicated. All times shall be as specified in Figure 2, Thermal cycle.

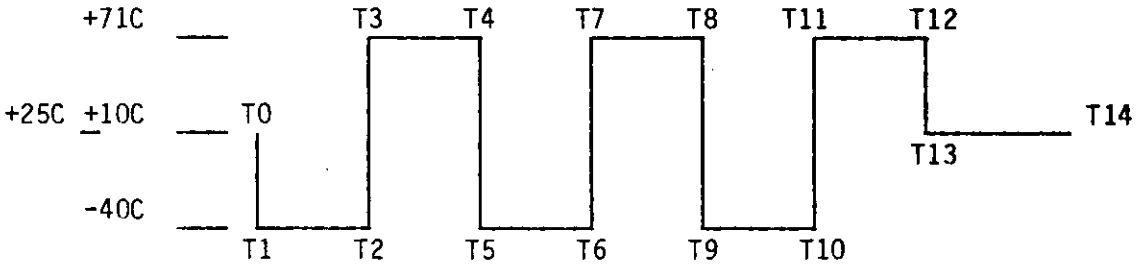
a. Prior to T0, test requirement 1 of Table V shall be conducted.

b. Between T9 and T10, test requirement 1 of Table V shall be conducted.

c. Between T10 and T11, test requirement 5 of Table V shall be conducted.

d. Between T11 and T12, test requirement 1 of Table V shall be conducted.

e. After T13, test requirement 1 of Table V shall be conducted.



T0 - T1 = room ambient to -40C, shall be not greater than 15 minutes.
T1 - T2 = -40C dwell time.
T2 - T3 = -40C to +71C, shall be not greater than 30 minutes.
T3 - T4 = +71C dwell time.
T4 - T5 = +71C to -40C, shall be not greater than 30 minutes.
T5 - T6 = -40C dwell time.
T6 - T7 = -40C to +71C, shall be not greater than 30 minutes.
T7 - T8 = +71C dwell time.
T8 - T9 = +71C to -40C, shall be not greater than 30 minutes.
T9 - T10 = -40C dwell time.
T10 - T11 = -40C to +71C, shall be not greater than 30 minutes.
T11 - T12 = +71C dwell time.
T12 - T13 = +71C to ambient, shall be not greater than 15 minutes.
T13 - T14 = room ambient.

FIGURE 2. Thermal cycle.

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4.8.1.2 Vibration conditioning test. The vibration conditioning test shall be in accordance with MIL-STD-810, Method 514.2. The sets shall be vibrated for 10 minutes in the axis perpendicular to the printed circuit boards and the longitudinal axes in accordance with Figure 514.2-4 of MIL-STD-810 using the test levels in Table VI, Test levels and frequencies. The following tests shall be performed at the indicated times:

a. Prior to vibration, test requirement 1 of Table V shall be conducted.

b. During each axial vibration, test requirement 2 of Table V shall be conducted, except during 4.9.1.1.1 only three input values shall be necessary. These shall be the low end point, mid scale, and high end point voltages.

c. After vibration has been completed, test requirement 1 of Table V shall be conducted.

4.8.2 Temperature-altitude test. The temperature-altitude test shall be conducted in accordance with MIL-STD-810, Method 504.1, Procedure 1, equipment catagory 7, except the nonoperating temperature range shall be -54C to +85C, the operating temperature range shall be -40C to +71C, the time at any test step shall be not greater than two hours, and steps 6 through 10, 12, and 13 shall be omitted. Steps 2, 11, and 14 shall be limited to one operating cycle. The following tests shall be performed as follows:

a. During step 2, test requirement 3 of Table V shall be conducted.

b. During step 3, test requirement 4 of Table V shall be conducted.

c. During step 4, test requirement 4 of Table V shall be conducted.

d. During step 5, test requirement 1 of Table V shall be conducted.

e. During step 11, test requirement 4 of Table V shall be conducted.

f. During step 14, test requirement 4 of Table V shall be conducted.

g. During step 15, test requirement 1 of Table V shall be conducted.

4.8.3 Shock test. A nonoperating shock test shall be conducted in accordance with MIL-STD-810, Method 516.2, Procedure IV, Figure 516.2-2, except peak value (A)G shall be 30 G and the nominal duration (D) shall be six milliseconds. The following tests shall be performed at the times specified:

a. Prior to shock test, test requirement 1 of Table V shall be conducted.

b. After shock test, test requirement 1 of Table V shall be conducted.

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TABLE V. Detailed performance test methods and procedures.

Test methods and procedures paragraph	Test requirements				
	1	2	3	4	5
4.9.1, 4.9.1.1 through 4.9.1.1.3.3	XXX		XXX	XXX	
4.9.1, 4.9.1.1 through 4.9.1.1.1.2		XXX			
4.9.1, 4.9.1.1, and 4.9.1.1.2		XXX			
4.9.1, 4.9.1.2 through 4.9.1.2.1.2	XXX	XXX	XXX	XXX	
4.9.1, 4.9.1.3 through 4.9.1.3.3	XXX	XXX	XXX	XXX	
4.9.1, 4.9.1.4 through 4.9.1.4.2.2	XXX		XXX	XXX	
4.9.1, 4.9.1.4 through 4.9.1.4.1.2		XXX			
4.9.1, 4.9.1.5 through 4.9.1.5.3	XXX	XXX	XXX	XXX	
4.9.1.6					XXX
4.9.2 through 4.9.2.2.2	XXX				
4.9.2, 4.9.2.2 through 4.9.2.2.2		XXX			
4.9.2 and 4.9.2.2.1			XXX		
4.9.2 and 4.9.2.2.2				XXX	
4.9.3 through 4.9.3.4	XXX		XXX	XXX	
4.9.3.1		XXX			XXX
4.9.3.2		XXX			XXX
4.9.4 through 4.9.4.1	XXX		XXX	XXX	
4.9.5	XXX		XXX	XXX	
4.9.6	XXX	XXX	XXX	XXX	

TABLE VI. Test levels and frequencies.

Test level	W_1 = 0.02 G^2/Hz
	W_2 = 0.04 G^2/Hz
Frequency	f_1 = 500 Hz
	f_2 = 1500 Hz

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4.8.4 Vibration test. Vibration testing shall be in accordance with MIL-STD-810, Method 514.2. The sets shall be vibrated for 10 minutes in each of three mutually perpendicular axes, one of which shall be longitudinal, in accordance with Figure 514.2-4 of MIL-STD-810, using the test levels and frequencies specified herein in Table VI, Test levels and frequencies. The following tests shall be performed at the times specified:

a. Prior to vibration test, requirement 1 of Table V shall be conducted.

b. During each axial vibration test, requirement 2 of Table V shall be conducted, except that during 4.9.1.1.1 only three input levels shall be required. These shall be the low end, mid scale, and high end voltages.

c. After vibration has been completed on all three axes, requirement 1 of Table V shall be conducted.

4.8.5 Humidity test. The set shall be tested in accordance with MIL-STD-810, Method 507.1, Procedure 1, except step 5 shall be for a total of five cycles with a total time of not less than 120 hours. The following tests shall be performed at the times specified:

a. Prior to humidity test, test requirement 1 of Table V shall be conducted.

b. During step 3 of the humidity test, test requirement 1 of Table V shall be conducted.

c. During step 6 of the humidity test, test requirement 1 of Table V shall be conducted.

d. After the humidity test, test requirement 1 of Table V shall be conducted.

4.9 Detailed performance test methods and procedures.

4.9.1 Signal conditioning tests. Unless otherwise specified herein, the tests specified in 4.9.1.1 through 4.9.1.5.3 shall be conducted after application of external power (see 3.6.1.1). Unless otherwise specified herein, all outputs shall be measured after demodulation by a telemetry receiver.

4.9.1.1 Analog channel tests. The signal characteristics of the analog channels shall be as specified in Table IX. The analog channels are Item 25 (A2) through Item 46 (A22). The voltages shall be applied to the points specified under Conn. pin no. through the source impedances specified under Source impedance. All analog inputs on P34 and P35 are referenced to P34-24. All analog inputs on P1 are referenced to P1-13.

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4.9.1.1.1 Accuracy and linearity test. Each channel shall be tested for channel accuracy and linearity using a five point test.

4.9.1.1.1.1 Analog channel input. The inputs for the five point test shall consist of the end point voltages, as specified in the Source voltage column, and one quarter scale, mid scale, and three quarter scale voltages relative to the end points.

4.9.1.1.1.2 Analog channel output. Each input shall produce the output as specified under Output count. The midscale output shall be equidistant from the end points. The one quarter and three quarter scale inputs shall produce outputs which shall be equidistant from their respective end points and midscale. The outputs shall be obtained by averaging the output of the test channel through not less than 20 consecutive frames of data. Each output shall be within ±3 bits of the specified Output count. The linearity of the channel shall be calculated and the maximum deviation shall be not greater than ± 2 bits from the best straight line.

4.9.1.1.2 Noise test. The inputs shall be as specified in 4.9.1.1.1.1. The outputs for each of the five input levels per channel shall be averaged through not less than 20 frames of data. The maximum and minimum count (see 6.4.2) for each level on each channel shall be within ±3 bits of the average value for that input level and channel.

4.9.1.1.3 Crosstalk test. Each analog channel shall be tested for crosstalk effects from the prior adjacent analog channel.

4.9.1.1.3.1 Crosstalk test channel inputs. Each analog channel under test shall be excited with 17 input voltages as specified in Table IX and Table VII. Table IX specifies the end point voltages under Source voltage and Table VII specifies the percent full scale difference relative to these voltages.

TABLE VII. Percentage of input range.

Input	Percentage	Input	Percentage
1	-2	10	52
2	0	11	74
3	2	12	76
4	18	13	78
5	20	14	80
6	22	15	98
7	24	16	100
8	28	17	102
9	50		

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4.9.1.1.3.2 Crosstalk alternate channel inputs. The alternate channel shall be the closest analog channel to appear prior to the first occurance of the test channel. The end point voltages specified in Table IX shall be applied to the alternate channel.

4.9.1.1.3.3 Crosstalk test method. Crosstalk shall be tested by the following:

a. Each input voltage from 4.9.1.1.3.1 shall be applied to the channel under test.

b. The low end point voltage shall be applied to the alternate channel for each test channel input. The test channel output count shall be averaged through not less than 20 frames for each test channel input voltage. The maximum and minimum output count for each word shall be recorded for the test channel.

c. The high end point voltage shall be applied to the alternate channel for each test channel input. The test channel output count shall be averaged through not less than 20 frames for each test channel input voltage. The maximum and minimum output count for each word shall be recorded for the test channel.

d. The average test channel output count for each test channel input voltage shall be the same percentage of the full scale output ±3 bits as the percentage of full scale input. The maximum and minimum of each output count shall be within ±3 bits of the average.

e. The test channel average output counts for high end point and low end point inputs shall differ by not greater than three bits.

f. The output count averages of the test channel shall be linear within ±2 bits.

4.9.1.2 Discrete channel tests. The signal characteristics of the discrete channels shall be as specified in Table IX. The discrete channels are Item 4 through Item 23 (D1-1 through D3-7). The voltages shall be applied to the points specified under Conn. pin no. through the source impedances specified under Source impedance. All discrete inputs on P34 and P35 are referenced to P34-24. All inputs on P19 are referenced to P1-13.

4.9.1.2.1 Discrete channel accuracy test. Each discrete channel shall be tested for accuracy using a two point test.

4.9.1.2.1.1 Discrete inputs. The inputs voltages for the two point accuracy test shall be as specified under Source voltage of Table IX.

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4.9.1.2.1.2 Discrete outputs. The outputs for the discrete channels shall be as specified under Output count of Table IX. The value of the output shall not change under constant input for not less than 20 consecutive full frames of data.

4.9.1.3 MBC tests. The signal characteristics of the MBC signals shall be as specified in Table IX. The MBC data signals are Item 53 (MBC A) through Item 56 (MBC D). The MBC control signals are Item 57 and 58. All signals are referenced to P34-24. The inputs shall be buffered using the devices listed under Source impedance in Table IX. The timing relationships between these signals shall be as specified in Figure 3. The cable length between the signal sources and the set shall be not greater than three feet.

4.9.1.3.1 MBC word location. The input shall consist of bursts of data every four milliseconds. The burst of data shall consist of 124 nibbles with each bit positioned as specified in Figure 3. The GATED CLOCK on P34-3 shall be adjusted so that t_L shall be equal to 400 nanoseconds and t_p shall be equal to 800 nanoseconds (see Figure 3). The bit pattern shall cause each MBC word to have the binary equivalent of the MBC word number shown in Table I. The value of each MBC word shall not change through 20 consecutive frames of data.

4.9.1.3.2 MBC word value. The input shall consist of bursts of data every four milliseconds. The burst of data shall consist of 124 nibbles. The GATED CLOCK shall be adjusted so that t_L shall be equal to 400 nanoseconds. Each nibble shall have a binary value:

a. equal to 0101. The bit pattern of each word in Table I shall have a binary value of 01010101. The value of each MBC word shall not change through 20 consecutive frames of data.

b. equal to 1010. The bit pattern of each word in Table I shall have a binary value of 10101010. The value of each MBC word shall not change through 20 consecutive frames of data.

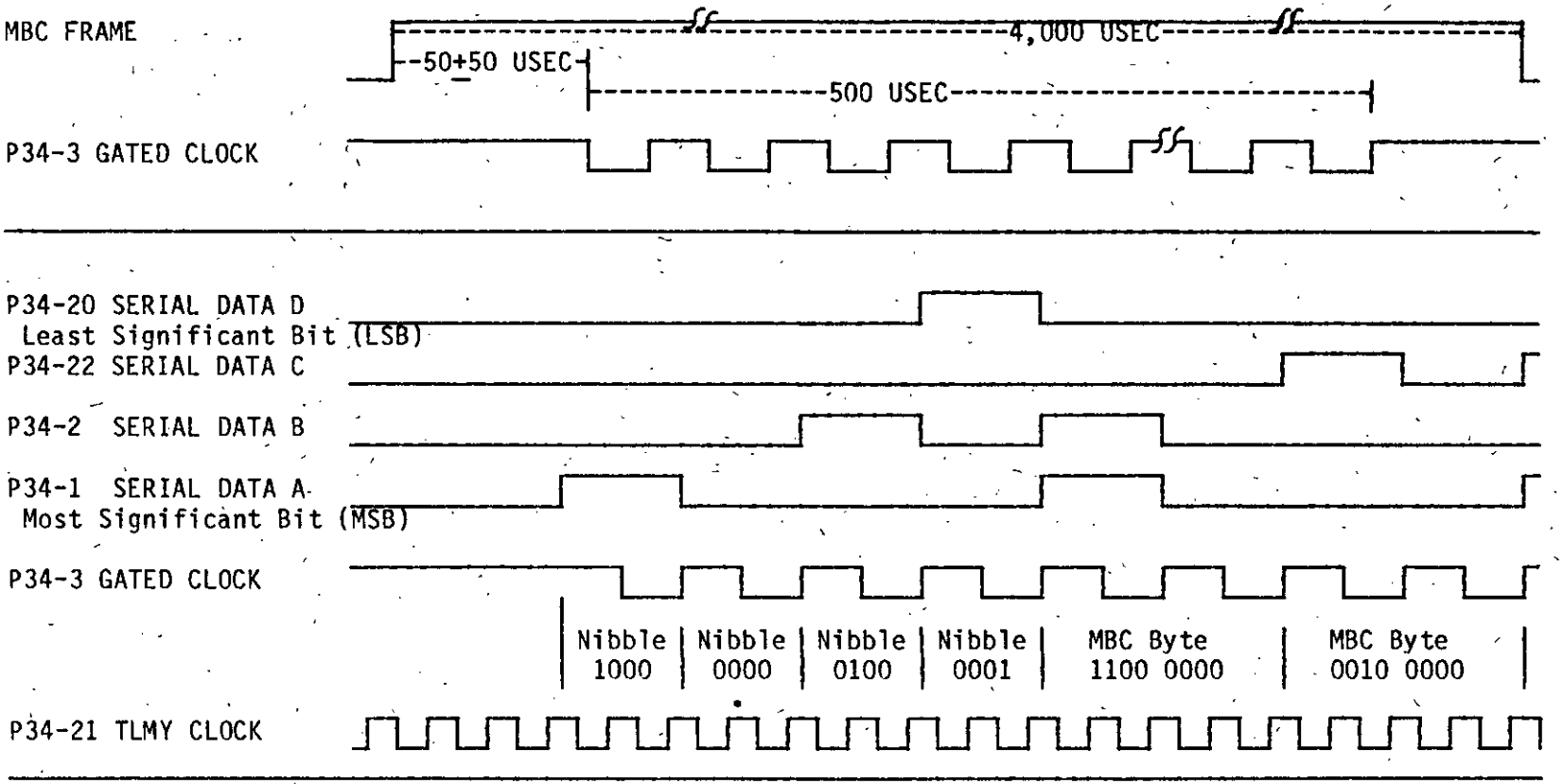
4.9.1.3.3 MBC clock variation tests. The tests of 4.9.1.3.2 shall be repeated two additional times. The GATED CLOCK shall have a value of T_L:

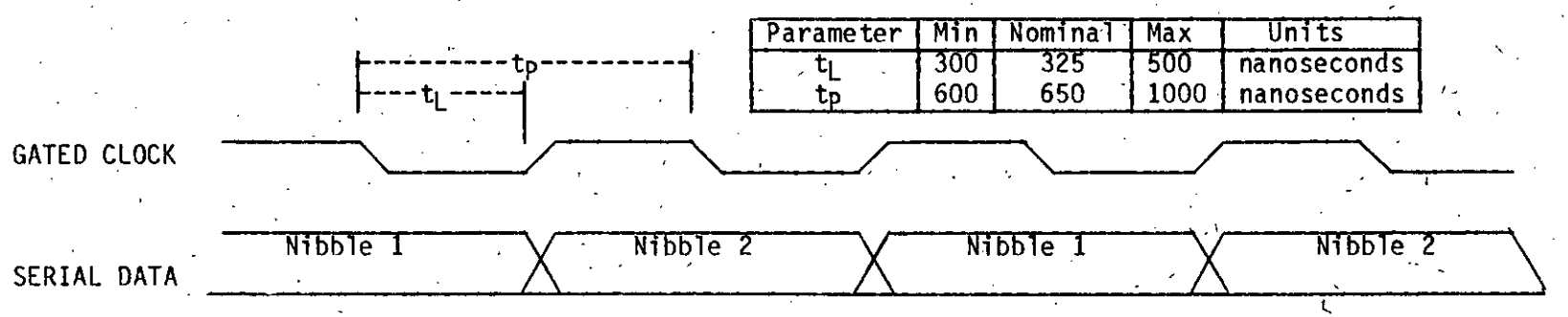
a. equal to 300 nanoseconds.

b. equal to 500 nanoseconds.

4.9.1.4 Video tests. The signal characteristics of the video and video control signals shall be as specified in Table IX. The video signals are Items 60, 62, 63, and 64. The control signal is Item 59. The control signal shall be applied to the point specified under Conn. pin no. through the source impedance specified under Source impedance. Item 59 is referenced to P34-24. Item 60 is referenced to P35-27. Items 62, 63, and 64 are coaxial connectors. All output measurements shall be made using the demodulated output of a telemetry receiver.

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Parameter	Min	Nominal	Max	Units
t_L	300	325	500	nanoseconds
t_p	600	650	1000	nanoseconds

FIGURE 3. MBC data and control signal characteristics.

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4.9.1.4.1 Injected Video test. Injected video shall be selected by applying 0.00 Vdc to P35-21.

4.9.1.4.1.1 Injected video input. The input to P35-9 shall be a sine wave as specified in Table IX, Item 60. The minimum input test frequencies shall be 10, 50, and 200 kHz.

4.9.1.4.1.2 Injected video output. The injected video output frequency shall be the same as the input frequency. The peak transmitter deviation shall be 155 kHz ± 10 kHz. All harmonic and spurious components shall be not greater than one percent.

4.9.1.4.2 Reconstructed video test. Reconstructed video shall be selected by applying 6.00 Vdc to P35-21.

4.9.1.4.2.1 Reconstructed video input. The inputs to P16 and P17 shall be sine waves as specified in Table IX and shall remain constant throughout the test. The input to P15 shall be a sine wave of five amplitudes at each of three frequencies. The amplitudes shall be -100, -90, -50, -10, and 0 decibels referenced to one milliwatt (dBm). The frequencies shall be Q+10, Q+50, and Q+200 kHz. The frequency Q shall be as specified on drawing 917AS6626.

4.9.1.4.2.2 Reconstructed video output. The reconstructed video output frequency shall be the same as the input frequency on P15 minus Q ± 200 Hz. The peak transmitter deviation shall be 155 kHz ± 10 kHz. All harmonic and spurious components shall be not greater than one percent.

4.9.1.5 Composite fuse tests. The characteristics of the composite fuse shall be as specified under Item 24 in Table IX and are divided into five groups,"a" through "e", each composed of four signals. All composite fuse inputs are referenced to P34-24.

4.9.1.5.1 Composite fuse inputs. The input voltages shall be as specified under Source voltage in Table IX. The voltages shall be applied to the points specified under Conn. pin no. through the source impedances specified under Source impedance. The inputs shall be applied in groups (see 4.9.1.5).

4.9.1.5.2 Composite fuse PCM output. The composite fuse PCM output shall be as specified under Output count in Table IX. The outputs shall increase monotonically and the difference in count between groups shall be not less than six bits.

4.9.1.5.3 Composite fuse tone burst outputs. The composite fuse tone burst outputs and transmitter deviations shall be as specified in Table II. Each group shall enable the single tone burst oscillator specified under Comments in Table IX.

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4.9.1.6 Input impedance test. The input impedance shall be measured and shall be in accordance with 3.6.2.6.

4.9.2 Operating power test. The characteristics of operating power shall be as specified in Table IX. External 400 Hz power includes Item 2 through Item 4. P34-26 is the reference for the external power. Missile battery power includes Item 21 through Item 23. P34-24 is the reference for missile battery power. The power shall be applied to the points specified under Conn. pin no. through the source impedances specified under Source impedance.

4.9.2.1 External power test.

4.9.2.1.1 External power low limit. The lowest voltage as specified under Source voltage in Table IX shall be applied.

4.9.2.1.2 External power high limit. The highest voltage as specified under Source voltage in Table IX shall be applied.

4.9.2.2 Missile battery power test. External power shall be applied prior to the application of battery power. External power shall be removed after missile battery power has been applied.

4.9.2.2.1 Missile battery power low limit. The lowest absolute value of the voltages as specified under Source voltage in Table IX shall be applied.

4.9.2.2.2 Missile battery power high limit. The highest absolute value of the voltages as specified under Source voltage in Table IX shall be applied.

4.9.3 Transmitter tests.

4.9.3.1 RF power output test. The RF transmitter incident power shall be not less than two watts under the following conditions:

a. The transmitter shall have a 50 ohm resistive load.

b. The transmitter load shall provide a VSWR of not less than 6.0:1 and the phase angle shall be varied from 0 degrees to 360 degrees.

4.9.3.2 RF carrier frequency test. The RF transmitter frequency shall be within ±0.003 percent of the assigned frequency under the following conditions:

a. The transmitter load shall be the antenna.

b. The transmitter shall have a 50 ohm resistive load.

c. The transmitter load shall provide a VSWR of not less than 6.0:1 and the phase angle shall be varied from 0 degrees to 360 degrees.

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4.9.3.3 Modulation test. The output of the set shall be monitored and the transmitter peak deviations shall be as specified in Table II.

4.9.3.4 Intermodulation and harmonic distortion test. The output of the set shall be monitored and all intermodulation and harmonic distortion shall be not less than 40 dB below the level of the injected video.

4.9.4 Transmitter disable test. The characteristics of the signals used to control the transmitter shall be as specified in Table IX. The signals are Item 49 (SEA SP COMMAND), Items 2 through 4 (400 Hz external power), and Items 21 through 23, (battery power). The voltages shall be applied to the points specified under Conn. pin no. through the source impedances specified under Source impedance. All inputs on P34 and P35 are referenced to P34-24 with the exception of Items 2 through 4 which are referenced to P34-26.

4.9.4.1 Transmitter disable inputs. All signal inputs to the set shall be disconnected prior to the application of Item 49 and Items 2 through 4. The tests shall be conducted by applying the control signals in the sequence specified in Table VIII.

4.9.5 Antenna test. The antenna VSWR shall be measured at a minimum of three frequencies: 2200, 2250, and 2300 MHz. The VSWR shall be not greater than 2.0:1.

4.9.6 VCO test. The deviation of the VCO by the PCM shall be measured and shall be ± 92 kHz with a tolerance of ± 10 kHz centered within ± 10 kHz of 450 kHz. The peak transmitter carrier deviation by the VCO shall be as specified in Table II.

4.9.7 Design and construction, and interchangeability. To determine conformance to 3.4 and 3.5.2, visual and dimensional inspections shall be performed to verify conformance with the dimensional and marking requirements of all drawings listed on DL1471AS100. Drawing requirements which cannot be verified after assembly shall be verified by use of the contractor's test data representing measurements recorded prior to assembly.

4.9.8 Workmanship. The set shall be visually inspected to ensure neatness, uniformity, and smoothness of coatings and paint; and absence of burrs, sharp edges, and foreign matter. Quality of electronic assembly shall be in accordance with WS-6536.

4.9.9 Electromagnetic interference. The set shall be tested for compliance with MIL-STD-461 and IRIG 106-80 in accordance with MIL-STD-462, Method CE03, and RE03. An EMI test plan and EMI test report shall be prepared by the contractor and shall be subject to acceptance by the procuring activity (see 6.2.2d and 6.2.2e).

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TABLE VIII. Transmitter disable test.

Test no.	SEA SP COMMAND	400 Hz power	Battery power	Set functional indication
1	-23.5 Vdc	Off	Off	Off
2	-23.5 Vdc	Change to On	Off	On for 205 ±5 seconds
3	-23.5 Vdc	Change to Off	Off	Off 1/
4	-23.5 Vdc	Change to On	Off	On for 205 ±5 seconds
5	-23.5 Vdc	On	Change to On	On continuous
6	-23.5 Vdc	Change to Off	On	On continuous
7	Change to -18.0 Vdc	Off	Change to Off	Off
8	-18.0 Vdc	Change to On	Off	On for 205 ±5 seconds
9	-18.0 Vdc	Change to Off	Off	Off 1/
10	-18.0 Vdc	Change to On	Off	On for 205 ±5 seconds
11	Change to +0.60 Vdc	Change to Off	Off	Off
12	+0.60 Vdc	Change to On	Off	On continuous
13	+0.60 Vdc	On	Change to On	On continuous
14	+0.60 Vdc	Change to Off	On	On continuous
15	+0.60 Vdc	Off	Change to Off	Off

1/ 400 Hz off for one second.

4.9.10 Failure classification. Failures shall be classified and reclassified in accordance with MIL-STD-2074 except that burn-in shall not apply. Failure reports shall be prepared by the contractor and shall be subject to the acceptance by the procuring activity (see 6.2.2j).

5. PACKAGING

5.1 Preservation, packaging, and packing. Unless otherwise specified in the contract or purchase order as level A or C (see 6.2.1d), the preservation, packaging, and packing shall be in accordance with Drawing 1471AS219.

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5.1.1 Level A.

5.1.1.1 Preservation and packaging. Preservation and packaging for level A shall be in accordance with MIL-P-116, Method III.

5.1.1.2 Packing. Packing for level A shall be in accordance with PPP-B-601, overseas type, style B, load type 1. The set shall be blocked, braced, cushioned, and anchored to prevent movement and damage to the set.

5.1.2 Level C.

5.1.2.1 Preservation and packaging. Preservation and packaging for level C shall be in accordance with MIL-P-116, Method III.

5.1.2.2 Packing. Packing for level C shall be acceptable by common carrier and shall ensure safe delivery at the lowest rate to the point of delivery. Packing shall comply with the carrier's rules and regulations applicable to the mode of transportation.

5.1.3 Spare parts. Unless otherwise specified in the contract or purchase order (see 6.2.1d), spare parts shall be preserved, packaged, and packed in accordance with MIL-P-116, Method III.

5.2 Marking. Marking for shipment shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Telemetric Data Transmitting Set AN/DKT-61(V) is intended for use in the AIM/RIM-7M missiles designated by the Government for testing and training exercises. The transmission of missile data associated with the tests and exercises will be accomplished by use of the set installed in place of the Missile Warhead Section.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Quantity of transmitting sets, AN/DKT-61(V), required.

c. Specific transmitting frequencies for each set ordered (see 3.5.4).

d. Level of preservation and packaging if shipping and storage container (Drawing 1471AS219)) is not used (see 5.1 and 5.1.3).

e. Quantity of preproduction sets for first article sample if different than 4.6.3 and 6.3.

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f. Disposition of sets (see 3.8 and 4.6.1).

g. Responsibility for inspection if different than 4.1.

h. Option to waive first article inspection (see 4.6).

i. Test system certification if other than specified (see 4.5.1.2).

j. Facility approved to perform first article and quality conformance inspections (see 4.6 and 4.7.1).

k. Inspection lot size if other than specified (see 4.7.2).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9(n)(2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

	Paragraph no.	Data requirement title	Applicable DID no.	Option
a.	3.3.1	Engineering Change Proposals (ECPs) and Requests for Deviations and Waivers (Long Form)	DI-E-2037	-------
b.	4.6 and 4.3.1	First Article Inspection Procedure	DI-T-4901	-------
c.	4.6.2 and 4.3.2	First Article Inspection Report	DI-T-4902	-------
d.	4.9.9	Electromagnetic Interference Test Plan	DI-R-7063	-------
e.	4.9.9 and 4.3.2	Electromagnetic Interference Test Report	DI-R-7062	-------
f.	4.7.3.1 and 4.3.2	Reports, Reliability Test and Demonstration	DI-R-7034	-------
g.	4.7.3 and 4.3.1	Procedures, Reliability Tests	DI-R-7035	-------
h.	4.7 and 4.3.1	Production/Acceptance Inspection Procedures	DI-T-4903	-------

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6.2.2 Data requirements. -Continued

Paragraph no.	Data requirement title	Applicable DID no.	Option
i. 4.7 and 4.3.2	Production Inspection Reports	DI-T-4904	-------
j. 3.3.1, 4.6.1, and 4.9.10	Report, Failed Item Analysis	DI-R-7039	-------

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer).

6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of two units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of first article sample (see 6.2.1e).

6.4 Definitions.

6.4.1 Satisfactory operation. Conforming to and meeting all requirements of this specification.

6.4.2 Count. The decimal equivalent of the binary value of a PCM data word.

6.4.3 Bit. The decimal equivalent of one least significant bit of a PCM data word.

6.4.4 Herein. In this specification.

6.4.5 Incident power. The product of the outgoing current and voltage traveling from the transmitter down a transmission line to a load.

Preparing activity:

Navy - AS

(Project 5821-N167)

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TABLE IX. Signal characteristics.

Item	Name	Conn. pin no	Source voltage	Source impedance 1/	Frame word	Bit no.	Output count	Freq. resp. 4/	Comments	
1	400HZ RTN	P34-26	NA	0 ohms	NA	NA	NA	NA	External power return	
2	PH 2, 113VRMS, 400HZ	P34-7	113±5 Vrms	0 ohms	NA	NA	NA	NA	External power	3 Phase 400±10 Hz
3	PH 3, 113VRMS, 400HZ	P34-25	113±5 Vrms	0 ohms	NA	NA	NA	NA	External power	
4 D1-3	PH 1, 113VRMS, 400HZ	P34-6	113±5 Vrms	0 ohms s	064	3	1 0	3/ 250 sps	External power 400 Hz present indication	1 present 0 absent
5 D1-1	INTERCEPT ARM	P35-32	0.00 Vdc +15.0 Vdc	100 kilohms	064	1	1 0	250 sps	Level detection, (See Item 24) +15.0 Vdc enables 3.0 kHz Tone Burst Oscillator	
6 D1-2	FUSE TRIG PULSE	P35-2	0.00 Vdc +5.50 Vdc	3.01 kilohms	064	2	1 0	250 sps	Level detection, (See Item 24) +5.50 Vdc enables 10.5 kHz Tone Burst Oscillator	
7 D1-5	FUSE DETECTION	P35-23	0.00 Vdc +14.0 Vdc	8.06 kilohms	064	5	1 0	250 sps	Level detection, (See Item 24), +14.0 Vdc enables 5.4 kHz Tone Burst Oscillator	
8 D1-6	FUSE OUT PULSE	P19-2	OPEN +26.5 Vdc	10.0 kilohms	064	6	1 0	250 sps	Level detection, (See Item 24), +26.5 Vdc enables 7.35 kHz Tone Burst Oscillator	
9	FUSE OUT PULSE RTN	P19-1	NA	0 ohms	NA	NA	NA	NA		
10 D1-7	LAUNCH +0.6 SEC	P1-28	2.50 Vdc +5.00 Vdc	100 kilohms	064	7	1 0	250 sps	Level detection	
11 D2-1	WB DETECT	P34-34	0.00 Vdc +6.01 Vdc	100 kilohms	123	1	1 0	250 sps	Level detection	
12 D2-2	AFC ENABLE	P34-33	0.00 Vdc +6.01 Vdc	100 kilohms	123	2	1 0	250 sps	Level detection	
13 D2-3	BOOST/ GLIDE	P1-12	0.00 Vdc +26.0 Vdc	20.0 kilohms	123	3	1 0	250 sps	Level detection	
14 D2-5	STOP SWEEP	P34-16	0.00 Vdc +6.01 Vdc	100 kilohms	123	5	1 0	250 sps	Level detection	

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TABLE IX. Signal characteristics. - Continued

Item	Name	Conn. pin no	Source voltage	Source impedance 1/	Frame word	Bit no.	Output count	Freq. resp. 4/	Comments
15 D2-6	LATCH COMMAND	P1-10	0.00 Vdc +26.0 Vdc	20.0 kilohms	123	1	6 0	250 sps	Level detection
16 D2-7	AFC F/H	P34-32	0.00 Vdc +6.01 Vdc	100 kilohms	123	7	1 0	250 sps	Level detection
17 D3-0	PHASE DET SWITCH	P34-17	+1.49 Vdc -1.49 Vdc	20.0 kilohms	013	0	1 0	250 sps	Level detection
18 D3-2	+5.9VDC	P34-11	+5.4 Vdc +4.1 Vdc	3.06 kilohms	013	2	1 0	250 sps	Level detection
19 D3-3	-8.8VDC	P34-30	-7.0 Vdc -7.6 Vdc	4.99 kilohms	013	3	1 0	250 sps	Level detection
20 D3-4	-26.5VDC	P34-10	-23.5 Vdc -25.5 Vdc	16.5 kilohms	013	4	1 0	250 sps	Level detection
21 D3-5	-20.6VDC	P34-29	-18.0 Vdc -19.0 Vdc	0 ohms	013	5	1 0	250 sps	Level detection
22 D3-6	+20.6VDC	P34-28	+19.0 Vdc +18.0 Vdc	0 ohms	013	6	1 0	250 sps	Level detection
23 D3-7	+26.5VDC	P34-9	+25.5 Vdc +23.5 Vdc	0 ohms	013	7	1 0	250 sps	Level detection
24 A1 a	COMPOSITE FUSE	P35-2 P35-23 P35-32 P19-2	0.00 Vdc 0.00 Vdc 0.00 Vdc 0.00 Vdc	See Item 5, 6, 7, and 8		NA	010 ±10	2000 sps	All Tone Burst Oscillators off
24 A1 b		P35-2 P35-23 P35-32 P19-2	0.00 Vdc 0.00 Vdc 0.00 Vdc +26.5 Vdc		001 017 033	NA	016 ±10		7.35 kHz ± 5% Tone Burst Oscillator enabled
24 A1 c		P35-2 P35-23 P35-32 P19-2	+5.50 Vdc 0.00 Vdc 0.00 Vdc 0.00 Vdc		049 065 081 097	NA	024 ±10		10.50 kHz ± 5% Tone Burst Oscillator enabled
24 A1 d		P35-2 P35-23 P35-32 P19-2	0.00 Vdc 0.00 Vdc +5.00 Vdc 0.00 Vdc		113	NA	040 ±10		3.00 kHz ± 5% Tone Burst Oscillator enabled
24 A1 e		P35-2 P35-23 P35-32 P19-2	0.00 Vdc +14.0 Vdc 0.00 Vdc 0.00 Vdc			NA	072 ±10		5.40 kHz ± 5% Tone Burst Oscillator enabled

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TABLE IX. Signal characteristics. - Continued

Item	Name	Conn. pin no	Source voltage	Source impedance 1/	Frame word	Bit no.	Output count	Freq. resp. 4/	Comments
25 A2	COMP TRK AGC	P35-30	-4.52 Vdc +4.52 Vdc	20.0 kilohms	010,042 074,106	NA	243 13	319 +32 Hz	
26	A2 GROUND REF	P34-4	NA	0 ohms	NA	NA	NA	NA	
27 A3	COMP ACQ AGC	P35-29	-4.53 Vdc +4.53 Vdc	20.0 kilohms	004,020 036,052 068,084 100,116	NA	243 13	614 +62 Hz	
28 A4	LOG LEVEL	P35-28	-3.81 Vdc +3.81 Vdc	20.0 kilohms	026,058 090,122	NA	243 13	316 +35 Hz	
29 A5	REAR COMP AGC	P34-31	-4.50 Vdc +4.50 Vdc	100 kilohms	080	NA	243 13	80 +8 Hz	
30 A6	REAR SWEEP	P34-15	-6.00 Vdc +6.00 Vdc	100 kilohms	005	NA	243 13	71 +7 Hz	
31 A7	SSLO POSITION	P34-37	+7.20 Vdc -7.20 Vdc	1,000 kilohms	002	NA	243 13	122 +12 Hz	
32 A8	DOPPLER ERROR WB	P35-12	-0.750 Vdc +0.750 Vdc	20.0 kilohms	011,043 075,107	NA	243 13	322 +33 Hz	
33 A9	DOPPLER ERROR NB	P35-31	-0.750 Vdc +0.750 Vdc	20.0 kilohms	045 109	NA	243 13	148 +15 Hz	
34 A10	PITCH RAD ERROR	P35-13	-5.42 Vdc +5.42 Vdc	20.0 kilohms	012,044 076,108	NA	243 13	355 +35 Hz	
35 A11	YAW RAD ERROR	P35-14	-5.42 Vdc +5.42 Vdc	20.0 kilohms	029,061 093,125	NA	243 13	355 +35 Hz	
36 A12	ANALOG CODING	P34-14	-1.81 Vdc +1.81 Vdc	100 kilohms	112	NA	243 13	85 +9 Hz	

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TABLE IX. Signal characteristics. - Continued

Item	Name	Conn. pin no	Source voltage	Source impedance 1/	Frame word	Bit no.	Output count	Freq. resp. 4/	Comments
37 A13	HEAD PITCH GYRO	P35-16	-11.25 Vdc +11.25 Vdc	20.0 kilohms	003	NA	243 13	82 ±8 Hz	
38 A14	HEAD YAW GYRO	P35-34	-11 25 Vdc +11.25 Vdc	20.0 kilohms	124	NA	243 13	82 ±9 Hz	
39 A15	WING 1 AND 3	P1-1	-8.33 Vdc +8.33 Vdc	20.0 kilohms	077	NA	243 13	80 ±8 Hz	
40 A16	WING 2	P1-2	-8.33 Vdc +8.33 Vdc	20.0 kilohms	016	NA	243 13	80 ±8 Hz	
41 A17	WING 4	P1-3	-8.33 Vdc +8.33 Vdc	20.0 kilohms	006	NA	243 13	80 ±8 Hz	
42 A18	PITCH GYRO	P1-5	-5.63 Vdc +5.63 Vdc	20.0 kilohms	008	NA	243 13	80 ±8 Hz	
43 A19	YAW GYRO	P1-6	-5.63 Vdc +5.63 Vdc	20.0 kilohms	009	NA	243 13	80 ±8 Hz	
44 A20	ROLL GYRO	P1-4	-4.50 Vdc +4.50 Vdc	20.0 kilohms	048	NA	243 13	78 ±8 Hz	
45 A21	IMAGE DOPPLER	P35-5	+9.00 Vdc -9.00 Vdc	7.50 kilohms	032 096	NA	243 13	157 ±18 Hz	
46 A22	ATTEN CONTROL	P35-22	+9.00 Vdc -9.00 Vdc	7.50 kilohms	007,023 039,055 071,087 103,119	NA	243 13	624 ±69 Hz	
47	SIM BAT BATTERY	P35-20	+26.5 Vdc	0 ohms	NA	NA	NA	NA	Test power source for the transmitter and voltage controlled oscillator
48	SIM BATTERY RTN	P35-25	NA	0 ohms	NA	NA	NA	NA	

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TABLE IX. Signal characteristics. - Continued

Item	Name	Conn. pin no	Source voltage	Source impedance 1/	Frame word	Bit no.	Output count	Freq. resp. 4/	Comments
49	SEA SP COMMAND	P35-33	+0.60 Vdc ±0.30 Vdc	21.0 kilohms	NA	NA	NA	NA	AIM-7M operation
			-20.5 Vdc ±3.00 Vdc	0 ohms					RIM-7M operation
50									This item not used
51	COM GND	P34-24	NA	0 ohms	NA	NA	NA	NA	
52	COM GND	P1-13	NA	0 ohms	NA	NA	NA	NA	
53 MBC A	SERIAL DATA A	P34-1	See MIL-M-38510/9 Device M38510/00905		See Figure 3, MBC data and control signal characteristics; and Table I, PCM frame configuration				
54 MBC B	SERIAL DATA B	P34-2							
55 MBC C	SERIAL DATA C	P34-22							
56 MBC D	SERIAL DATA D	P34-20							
57	TLMY CLOCK	P34-21	See MIL-M-38510/163 Device M38510/16304						
58	GATED CLOCK	P34-3							
59	LAUNCH SWITCH	P35-21	+6.00 Vdc 0.00 Vdc	20.0 kilohms	NA	NA	NA	NA	Controls switching between injected and reconstructed video. 0.00 Vdc=injected video +6.00 Vdc=reconstructed video
60	INJECTED VIDEO	P35-9	60±5 mVrms	20.0 kilohms	NA	NA	NA	NA	Frequency from 10 kHz to 200 kHz

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TABLE IX. Signal characteristics. - Continued

Item	Name	Conn. pin no	Source voltage	Source impedance 1/	Frame word	Bit no.	Output count	Freq. resp. 4/	Comments
61	SHIELD RTN	P35-27	NA	0 ohms	NA	NA	NA	NA	Shield return for injected video P35-9
62	IF	P15	0 dBm TO -105 dBm	50 ohms	NA	NA	NA	NA	Frequency from Q + 10 kHz 2/ to Q + 200 kHz
63	LOW FREQ REF	P16	100±10 mVrms	50 ohms	NA	NA	NA	NA	Frequency of 3,420,000 ± 100 Hz
64	HIGH FREQ REF	P17	100±10 mVrms	50 ohms	NA	NA	NA	NA	Frequency of Q + 3,420,000 ± 100 Hz 2/

1/ All Source impedance shall be ± one percent, with the exception of 0 ohms which shall be nominal.
2/ Q shall be as specified on drawing 917AS6626.
3/ Samples per second (sps).
4/ Frequency response.

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